

Step 1: Install Required Libraries

```
!pip install nltk spacy
```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Requirement already satisfied: spacy in /usr/local/lib/python3.12/dist-packages (3.8.14)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk) (8.3.1)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from nltk) (1.5.3)
Requirement already satisfied: regex<=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk) (2025.11.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk) (4.67.3)
Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.12)
Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.5)
Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.15)
Requirement already satisfied: cymem<2.1.0,>=2.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.13)
Requirement already satisfied: preshed<3.1.0,>=3.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.13)
Requirement already satisfied: thinc<8.4.0,>=8.3.12 in /usr/local/lib/python3.12/dist-packages (from spacy) (8.3.13)
Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.1.3)
Requirement already satisfied: srsly<3.0.0,>=2.5.3 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.5.3)
Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.10)
Requirement already satisfied: weasel<2.0.0,>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.0)
Requirement already satisfied: confection<2.0.0,>=1.3.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.3.3)
Requirement already satisfied: typer<1.0.0,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.24.1)
Requirement already satisfied: numpy>=1.19.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.2)
Requirement already satisfied: requests<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.32.4)
Requirement already satisfied: pydantic<3.0.0,>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.12.3)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.1.6)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from spacy) (75.2.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (26.0)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.0.0->sp)
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.0.0->sp)
Requirement already satisfied: typing-extensions>=4.14.1 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.0.0)
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.0.0)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spacy) (3.10.1)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spac)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spac)
Requirement already satisfied: blis<1.4.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.12->spacy) (1.3.0)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer<1.0.0,>=0.3.0->spacy) (1.5.4)
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer<1.0.0,>=0.3.0->spacy) (13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer<1.0.0,>=0.3.0->spacy) (0.0.2)
Requirement already satisfied: cloudpathlib>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from weasel<2.0.0,>=1.0.0->spacy) (0.19.0)
Requirement already satisfied: smart-open>=5.2.1 in /usr/local/lib/python3.12/dist-packages (from weasel<2.0.0,>=1.0.0->spacy) (7.0.5)
Requirement already satisfied: httpx>=0.24.0 in /usr/local/lib/python3.12/dist-packages (from weasel<2.0.0,>=1.0.0->spacy) (0.28.1)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from Jinja2->spacy) (3.0.3)
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx>=0.24.0->weasel<2.0.0,>=1.0.0->spacy) (4.6.2)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx>=0.24.0->weasel<2.0.0,>=1.0.0)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx>=0.24.0->weasel<2.0.0,>=1.0.0)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer<1.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer<1.0.0)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart-open>=5.2.1->weasel<2.0.0,>=1.0.0->sp)
Requirement already satisfied: mdurl~0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3.0->typer<1.0.0)
```

Step 2: Import Libraries

```
import nltk
import spacy
```

Step 3: Download Required Data (NLTK + spaCy)

```
nltk.download('punkt')
```

```
!python -m spacy download en_core_web_sm
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Unzipping tokenizers/punkt.zip.
Collecting en-core-web-sm==3.8.0
  Downloading https://github.com/explosion/spacy-models/releases/download/en\_core\_web\_sm-3.8.0/en\_core\_web\_sm-3.8.0-py3-none-any
12.8/12.8 MB 63.2 MB/s eta 0:00:00
✓ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in order to load all the package's dependencies. You can do this by selecting the 'Restart kernel' or 'Restart runtime' option.
```

Step 4: Load spaCy Model

```
nlp = spacy.load("en_core_web_sm")
```

Step 6: Add Your Text

```
text = """Artificial Intelligence is transforming the world.  
It helps in automation, decision making, and solving complex problems."""
```

Step 7: Convert Text to Lowercase

```
lower_text = text.lower()  
print("Lowercase Text:\n", lower_text)
```

```
Lowercase Text:  
artificial intelligence is transforming the world.  
it helps in automation, decision making, and solving complex problems.
```

Step 8: Count Number of Words

```
words = lower_text.split()  
word_count = len(words)  
  
print("Number of words:", word_count)
```

```
Number of words: 16
```

Step 9: Tokenization using NLTK

```
from nltk.tokenize import word_tokenize  
nltk.download('punkt_tab')  
tokens = word_tokenize(lower_text)  
print("Tokens:", tokens)
```

```
Tokens: ['artificial', 'intelligence', 'is', 'transforming', 'the', 'world', '.', 'it', 'helps', 'in', 'automation', ',', 'decis  
[nltk_data] Downloading package punkt_tab to /root/nltk_data...  
[nltk_data] Package punkt_tab is already up-to-date!
```